

Lab 1 — Algorithms and coding: three programs

TC1037 Algorithmic Thinking · Group 510 · Friday 25 September 2026,
16:30–17:30, Room 4206

Why this lab looks like this

From next week every one of the 13 groups runs **one** version of each lab (department decision, 22 Sep), so the TAs prepare one lab instead of six. This week we adopt the national exercises already: for each problem, **first the algorithm, then the program, then tests** — which is the method of this course anyway.

Running order

	Minutes	What
1	20	Initial prior-knowledge diagnostic — administered by the professor at 16:30. No built-in timer: the professor calls time at 20 minutes. You need your laptop for this part.
2	3	Settle. Everyone opens Thonny ; whoever does not have it pairs with a neighbour and installs during exercise 2. The professor opens the first levelling lesson on the projector for one minute.

	Minutes	What
3	15	Exercise 1 — together. Algorithm on paper (3 min, everyone), then Luis types the program on the projector line by line and students type along.
4	19	Exercises 2 and 3 — individual. Algorithm on paper first, then the program, then the tests.
5	3	Hand in / submit.

Bring **laptop, notebook and pencil** and the A_Matricula.docx algorithm template (printed Tuesday; blank paper is fine).

The rule for all three exercises

For each problem, in this order:

1. **The algorithm** — on paper. INPUT: line, OUTPUT: line, numbered steps. Written *before* any code. This is the part that is graded as the work; the program is its transcription.
2. **The program** — in Thonny. Start every file the same way: **line 1, your name in a # comment; then the algorithm inside a "" multi-line comment; then the code.**
3. **The tests** — run it with the values given and write, under the algorithm, the values you used and what the program printed. A program that runs is not a program that is correct.

Exercise 1 — Area of a triangle (*together, 15 min*)

Read the base and the height of a triangle from the user; display its area.

Area = base × height / 2

Tests: base 10, height 5 → 25.0. Base 3, height 4 → 6.0. Save as A_Matricula_1.py.

Exercise 2 — Dollars to pesos (*individual*)

Read an amount in US dollars and today's exchange rate (pesos per dollar); display the equivalent in pesos.

Tests: 100 dollars at 18.50 → 1850.0. 20 at 17.80 → 356.0. Save as A_Matricula_2.py.

Exercise 3 — Fahrenheit to Celsius (*individual*)

Read a temperature in degrees Fahrenheit; display it in degrees Celsius.

$$C = 5 / 9 \times (F - 32)$$

Tests: 212 → 100.0. 32 → 0.0. 98.6 → 37.0. If you wrote $5 / 9 * F - 32$, check whether you were lucky or right — and be able to say which. Save as A_Matricula_3.py.

Finished all three? Swap your Exercise 2 algorithm with a neighbour and execute theirs literally for 0, -3 and ho!a. Write down, in one line per input, what their steps actually produce. Hand that back with your name on it.

Submission

Two parts:

- **In the room, before you leave:** your three handwritten algorithms (with the test values and results written underneath) — to Luis. He records attendance and completion.
- **On Canvas, by 23:59 tonight** — *Laboratory 1*: one PDF or photo of the three algorithms, and the .py files you finished (A_Matricula_1.py, _2.py, _3.py). If you did not finish a program in the lab, finish it after tonight's session — the lecture at 17:35 covers exactly what these programs use — and upload it then.

An upload with code but no algorithm is **incomplete** and is graded as such. Programs are individual work: the triangle one was typed together, the other two are yours. See *Using AI in this course*.
